

Simulating Quantum Field Fluctuations in Minkowski Space

A report submitted to the University of Waterloo in
partial fulfillment of the requirements
for PHYS 785

By

Cristiano Sampaio

Abstract

This project aims to successfully plot quantum field fluctuations of the Klein-Gordon field and analyze the impacts of various parameters. This report begins with the Klein-Gordon equation as well as the commutation relations for $\hat{\phi}$, which were manipulated to provide new relations for the Fourier coefficients ϕ_n to obtain decoupled harmonic oscillators and ultimately their wave functionals. The amplitudes of ϕ_n were calculated and ultimately used to plot fluctuations of $\phi(x, y, z)$ for a given time slice.

The report continues to analyze various contour plots of $\phi(x, y)$ and determines how manipulating certain parameters such as the mass of the field, number of modes and dimensions used impacts the fluctuation plots. It was determined that increasing the mass of the field decreased the amplitude of the fluctuations of the field and that increasing the number of modes $\hat{\phi}_{\vec{n}}$ that are considered increased the amplitudes for the field fluctuations. More research is required to analyze excited states, as only a few number of particles are considered for some of the calculations.

Introduction/Motivation

This report will focus on obtaining quantum fluctuations of fields with plots for Minkowski space. Although in lectures, we have gone through the rigour and theory needed to obtain necessary functions for this task, the transformations used in this report differ considerably, as the Fourier sine transformation is used after infrared regularization. This changes our commutation relations and hermiticity condition considerably. Despite this, most of the intuition used in lectures still holds with certain caveats, which will be discussed in the Theory section of this report. This report aims to visualize how different fields and states can manipulate the fluctuations of said fields. Using this understanding for Minkowski space, we could hopefully further extend this into other spacetimes such as FRW spacetimes. Lastly, natural units will be used such that $\hbar = 1$.

Theory

Quantizing the Equations

In Minkowski space, we recall the Klein-Gordon equation is of the form:

$$(\partial_t^2 - \Delta + m^2)\hat{\phi}(x, t) = 0 \quad (1)$$

Where Δ is the Laplacian and m is the mass of the field. The commutation relations and hermiticity condition for the $\hat{\phi}(x, t)$ field are as follows:

$$[\hat{\phi}(\vec{x}, t), \hat{\phi}(\vec{x}', t)] = 0, [\hat{\phi}(\vec{x}, t), \hat{\pi}(\vec{x}', t)] = i\hbar\delta^3(\vec{x} - \vec{x}'), [\hat{\pi}(\vec{x}, t), \hat{\pi}(\vec{x}', t)] = 0, \quad (2)$$

where $\hat{\phi}(x, t) = \hat{\pi}(x, t)$. To simplify the procedure, we apply an infrared regularization in which we restrain our system to a box $[0, L] \times [0, L] \times [0, L]$ of arbitrary length L with the Dirichlet boundary conditions such that $\phi(\text{boundary}) = 0$. With this regularization as well as the discrete Fourier sine transformation, we have that in one dimension:

$$\hat{\phi}(x, t) = \sum_{n=1}^{\infty} \hat{\phi}_n(t) \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) \quad \text{where} \quad \hat{\phi}_n(t) = \int_0^L \hat{\phi}(x, t) \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) dx \quad (3)$$

Therefore, in three dimensions, the Fourier sine transformation is as follows:

$$\hat{\phi}(\vec{x}, t) = \sum_{n_1, n_2, n_3=1}^{\infty} \hat{\phi}_{\vec{n}}(t) \left(\frac{2}{L}\right)^{3/2} \sin\left(\frac{n_1\pi x}{L}\right) \sin\left(\frac{n_2\pi y}{L}\right) \sin\left(\frac{n_3\pi z}{L}\right) \quad (4)$$

$$\hat{\phi}_{\vec{n}}(t) = \int_0^L \hat{\phi}(\vec{x}, t) \left(\frac{2}{L}\right)^{3/2} \sin\left(\frac{n_1\pi x}{L}\right) \sin\left(\frac{n_2\pi y}{L}\right) \sin\left(\frac{n_3\pi z}{L}\right) dx^3 \quad (5)$$

Where $\vec{n} = (n_1, n_2, n_3)$ is the field's $\vec{n}$ -mode.

Using these transformations as well as the commutation relations and hermiticity condition for $\phi(x, t)$, new conditions and relations will be obtained for the coefficients $\phi_n(t)$ to ultimately obtain their probability distributions.

Beginning with the Klein-Gordon equation, by substituting in the Fourier transformation, equation (4) into (1) and evaluating the derivatives. We note that the time derivative can be taken into the summation directly to $\phi_n(t)$ as no other terms are time dependent.

$$\left(\frac{2}{L}\right)^{3/2} \sum_{n_1, n_2, n_3=1}^{\infty} \sin\left(\frac{n_1\pi x}{L}\right) \sin\left(\frac{n_2\pi y}{L}\right) \sin\left(\frac{n_3\pi z}{L}\right) (\partial_t^2 + \frac{\pi^2}{L^2} |\vec{n}|^2 + m^2) \hat{\phi}_{\vec{n}}(t) = 0,$$

We next multiply both sides by the sine function product and integrate over the regions of the box. Using the orthogonality condition of the sine function,

$$\frac{2}{L} \int_0^L \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{n'\pi x}{L}\right) dx = \delta_{n, n'},$$

We then obtain through simplification:

$$(\partial_t^2 + \frac{\pi^2}{L^2} |\vec{n}|^2 + m^2) \hat{\phi}_{\vec{n}}(t) = 0 \quad (6)$$

These are decoupled harmonic oscillators with frequencies $\omega_k^2 = k^2 + m^2$, where $k^2 = \frac{\pi^2}{L^2} |\vec{n}|^2$.

Next, the commutation relations for $\phi_n(t)$ are obtained. Through a simple Fourier inverse transformation, it is clear to see that $\hat{\phi}(x, t) = \hat{\pi}(x, t)$. Using this, we obtain:

$$[\hat{\phi}_{\vec{n}}(t), \hat{\pi}_{\vec{n}'}(t)] = \int_0^L \left(\frac{2}{L}\right)^3 \sin\left(\frac{\vec{n}\pi\vec{x}}{L}\right) dx^3 \int_0^L \sin\left(\frac{\vec{n}'\pi\vec{x}'}{L}\right) dx'^3 [\hat{\phi}(\vec{x}, t), \hat{\pi}(\vec{x}', t)]$$

Where $\sin\left(\frac{\vec{n}\pi\vec{x}}{L}\right) = \sin\left(\frac{n_1\pi x}{L}\right)\sin\left(\frac{n_2\pi y}{L}\right)\sin\left(\frac{n_3\pi z}{L}\right)$. Using (2), the commutator evaluates to the Dirac delta $\delta^3(\vec{x} - \vec{x}')$, which then eliminates one of the triple integrals through the sifting property. This leaves us with

$$[\hat{\phi}_{\vec{n}}(t), \hat{\pi}_{\vec{n}'}(t)] = \int_0^L \left(\frac{2}{L}\right)^3 \sin\left(\frac{\vec{n}\pi\vec{x}}{L}\right) \left(\frac{\vec{n}'\pi\vec{x}}{L}\right) dx^3$$

Using the orthogonality relation, we end up with our solution:

$$[\hat{\phi}_{\vec{n}}(t), \hat{\pi}_{\vec{n}'}(t)] = i\delta_{\vec{n}, -\vec{n}'} \quad (7)$$

This commutation relation is of great significance, as the commutation relations differ from the example in Lecture 3 where instead the same relation is $\delta_{\vec{n}, -\vec{n}'}$, and the coefficients $\hat{\phi}_{\vec{n}}(t)$ are not Hermitian. It is easy to see in this case that $\hat{\phi}_{\vec{n}}(t)$ is indeed Hermitian as they are real-valued. By choosing to transform through the Fourier sine transformation as opposed to the total Fourier transformation, we have omitted the imaginary part that would result in $\hat{\phi}_{\vec{n}}(t)$ being non-Hermitian. As for the other relations in (2), we receive identical relations to those of Lecture 3 trivially, where:

$$[\hat{\phi}_{\vec{n}}(t), \hat{\phi}_{\vec{n}'}(t)] = [\hat{\pi}_{\vec{n}}(t), \hat{\pi}_{\vec{n}'}(t)] = 0 \quad (8)$$

Solving for the Amplitudes $\phi_{\vec{n}}$

The technique used to solve for the various amplitudes of $\phi_{\vec{n}}$ follows Lecture 4, where we are now in the Schrodinger picture, where the observables no longer evolve in time but the states do. We now need to solve for the wave functional $\psi[\check{\phi}, t]$, (where $\check{\phi}$ is the Fourier transform of ϕ and we assume they are in the joint ground state) however first we must find a representation of relation (7). We consider the ansatz:

$$\hat{\phi}_{\vec{n}} \cdot \psi[\check{\phi}, t] = \phi_{\vec{n}} \psi[\check{\phi}, t] \text{ and } \hat{\pi}_{\vec{n}} \cdot \psi[\check{\phi}, t] = -i \frac{\partial}{\partial \phi_{\vec{n}}} \psi[\check{\phi}, t] \quad (9)$$

Which is verified below:

$$\begin{aligned}
[\hat{\phi}_{\vec{n}}, \hat{\pi}_{\vec{n}}, \cdot] \cdot \psi[\check{\phi}, t] &= (\hat{\phi}_{\vec{n}} \hat{\pi}_{\vec{n}}, - \hat{\pi}_{\vec{n}}, \hat{\phi}_{\vec{n}}) \cdot \psi[\check{\phi}, t] \\
&= -i \phi_{\vec{n}} \frac{\partial}{\partial \phi_{\vec{n}}} \psi[\check{\phi}, t] + i \frac{\partial}{\partial \phi_{\vec{n}}} (\phi_{\vec{n}} \psi[\check{\phi}, t]), \text{ using the chain rule} \\
&= i \delta_{\vec{n}, \vec{n}}, \psi[\check{\phi}, t]
\end{aligned}$$

Using the Hamiltonian which resembles the one used in lectures, for a harmonic oscillator,

$$\hat{H} = \sum_{n_1, n_2, n_3=1} \hat{H}_{\vec{n}}, \text{ where } \hat{H}_{\vec{n}} = \frac{1}{2} (\hat{\pi}_{\vec{n}}^2 + \omega_k^2 \hat{\phi}_{\vec{n}}^2) \quad (10)$$

Using the representation (9) and the Hamiltonian (10), we can rewrite the Schrodinger equation as follows:

$$\begin{aligned}
i \partial_t \psi[\check{\phi}, t] &= \hat{H} \psi[\check{\phi}, t] \\
i \partial_t \psi[\check{\phi}, t] &= \sum_{n_1, n_2, n_3=1} \frac{1}{2} \left(-\frac{\partial^2}{\partial \phi_{\vec{n}}^2} + \omega_k^2 \phi_{\vec{n}}^2 \right) \psi[\check{\phi}, t] \quad (11)
\end{aligned}$$

We see from (11) that the ansatz holds:

$$\psi[\check{\phi}, t] = N e^{-\sum_{n_1, n_2, n_3=1} \frac{1}{2} (\omega_k \phi_{\vec{n}}^2 - i \omega_k t)} \quad (12)$$

We want the probability amplitude density of the wave functional, thus we eliminate the phase to obtain:

$$|\psi[\check{\phi}, t]|^2 = N^2 e^{-\sum_{n_1, n_2, n_3=1} (\omega_k \phi_{\vec{n}}^2)}, \quad (13)$$

where N is some normalization constant. Now, we notice we can further simplify the probability density as a product of exponentials to obtain:

$$|\psi[\check{\phi}, t]|^2 = N^2 \prod_{\vec{n}} e^{-(\omega_k \phi_{\vec{n}}^2)} \quad (14)$$

We then therefore see that for a particular mode,

$$|\psi[\widehat{\phi}_{\vec{n}}, t]|^2 = N_{\vec{n}}^2 e^{-(\omega_k \phi_{\vec{n}}^2)} \quad (15)$$

Where $N_{\vec{n}}^2 = \sqrt{\frac{\omega_k}{\pi}}$, where $k^2 = \frac{\pi^2}{L^2} |\vec{n}|^2$. This agrees with our intuition that in the ground state, the probability distribution will resemble a Gaussian. With these probability distributions for each

mode, we can obtain the amplitudes of $\phi_{\bar{n}}$ and using (3), through the Fourier transformation we can sum these various modes to obtain a plot of $\hat{\phi}(x)$.

Excitation Case

For the case in which some of the modes are excited and are no longer in the ground state, the wave functional (14) is then multiplied by the Hermite polynomial $H_m(\sqrt{\omega_k}x)$, where n denoted the level of excitation. This can be verified with a similar approach to the ground state such that:

$$|\psi[\check{\phi}, t]|^2 = N^2 e^{-\sum_{n_1, n_2, n_3=1} (\omega_k \phi_{\bar{n}}^2)} (H_m(\sqrt{\omega_k}x))^2 \quad (16)$$

This new general function results in the probability distribution spreading out over various peaks. As opposed to the distribution being Gaussian, the new distribution will have 2 smaller peaks for larger amplitudes for the case of $m=1$. This in turn increases the likelihood of obtaining a larger $\phi_{\bar{n}}$, as shown in Figure 1 below.

Method

Determining the Amplitudes for Each Mode

To calculate each mode's respective amplitude, a plot of (15) is produced and a box is constructed to contain most of the probability density function, with box height equal to the maximum of (15) and the width determined through the Empirical Rule, which states that 99.7% of data observed following a normal distribution with mean 0 will be contained within the interval $[-3\sigma, 3\sigma]$. Note however, that the Empirical Rule only applies for unimodal distributions and thus will not succeed for the excited state distributions, which will be discussed later. After the construction of the box, amplitudes for $\phi_{\bar{n}}$ and $|\psi[\widehat{\phi}_{\bar{n}}, t]|^2$ are uniformly chosen until one of the ordered pairs lies directly underneath the distribution. This coordinate's $\phi_{\bar{n}}$ will be chosen to represent the amplitude for that given mode. An example of this process is shown in Figure 1 below.

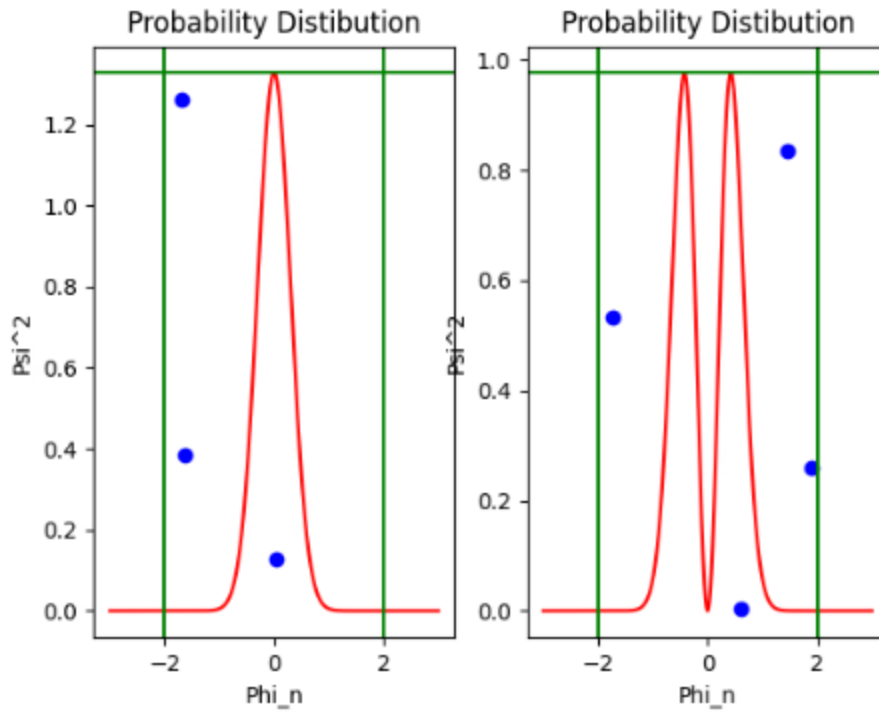


Figure 1: Probability distribution function for an arbitrary $\phi_{\vec{n}}$. Green lines indicate the boxed restraints placed for sampling. Illustrations of ground state probability distribution, $m=0$, (Left) and excited probability distribution, $m=1$ (Right).

Obtaining the Probability

Through the methods described above, after obtaining these amplitudes for various modes, we use the following Fourier transformation:

$$\phi(x, y, z) = \sum_{n_1, n_2, n_3=1}^{\infty} \phi_{\vec{n}} \left(\frac{2}{L}\right)^{3/2} \sin\left(\frac{n_1 \pi x}{L}\right) \sin\left(\frac{n_2 \pi y}{L}\right) \sin\left(\frac{n_3 \pi z}{L}\right) \quad (17)$$

Using this general formula, we will approximate these fluctuations by considering only a finite number of modes. With this, we can plot contour lines of these fluctuations by looking at slices of this function by fixing one of the parameters (example: $z=a, a \in [0, L]$)

The code I constructed that was used for this project can be found here:

<https://github.com/CSampaio101/QFT-Project.git>

Results

Numerous plots were obtained from (17) for varying parameters. The motivation for this was to determine how each parameter can impact the fluctuation plots. Significant changes in parameters will be noted, however $L=10$ for many plots. Furthermore, although it may not be clear through the figures below, the Dirichlet conditions were confirmed to be satisfied.

Mass

As shown in Figure 2, the amplitude of the fluctuations decreases with an increase in mass. This can also be seen mathematically, as we recall that (15) is dependent on $\omega_k = \sqrt{k^2 + m^2}$, and thus as the mass increases, the frequency of each decoupled harmonic oscillator increases, ultimately lowering the amplitudes of each $\phi_{\vec{n}}$. This can be also seen through the probability distribution of the wave functionals through Figure 1; as we increase the frequency, we are narrowing the spike of the Gaussian and reducing the likelihood for greater amplitudes.

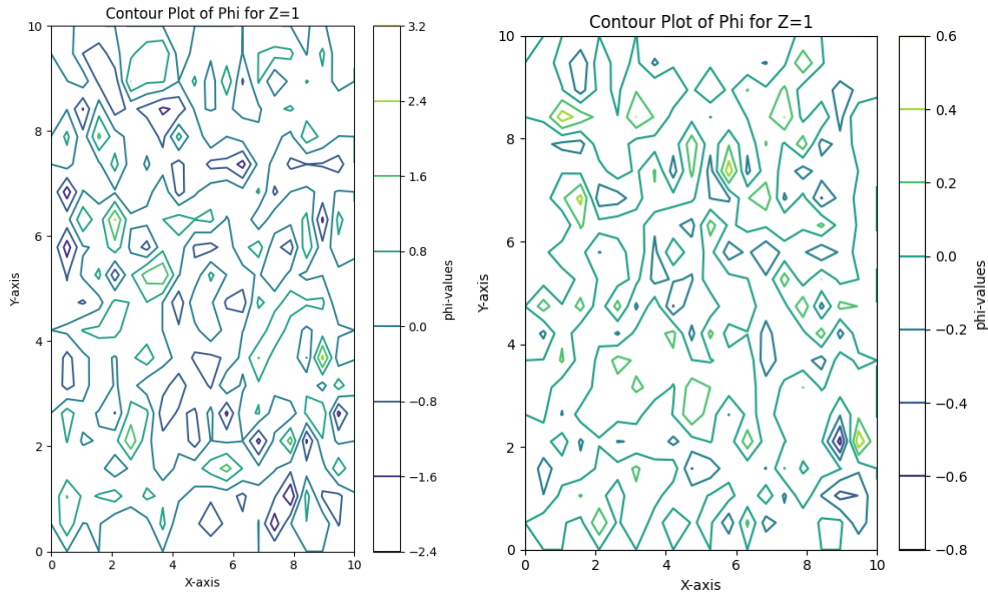


Figure 2: Plots of $\phi(x, y, 1)$ with a mass $m=0$ (Left) and a nonzero mass $m=100$ (Right). The number of modes considered was 20^3 (20 choices for each dimension).

Number of modes

As we increase the number of $\phi_{\vec{n}}$ used to approximate $\phi(x, y, z)$, we approach a more accurate representation of the field fluctuations. These plots were obtained by starting with $\vec{n} = (1,1,1)$ and checking every possible permutation until a desired amount of $\phi_{\vec{n}}$ are considered. As illustrated in Figure 3, the contour plots vary drastically. Contrary to the mass parameter, the amplitudes for the fluctuations increase with the number of modes. This is expected, as we are increasing the number of terms being considered which would increase the maximum and minimum of the function; although we may be adding more positive and negative terms, we expect the fluctuations to get larger. Of more significance, however, is the pattern of the fluctuations, as the amount of fluctuations in the region of the box increases. When considering only 27 modes, rings appeared for the contour lines, however as the number of modes increased, the contour lines became more chaotic and resembled more of what one would expect for the fluctuations in the ground state.

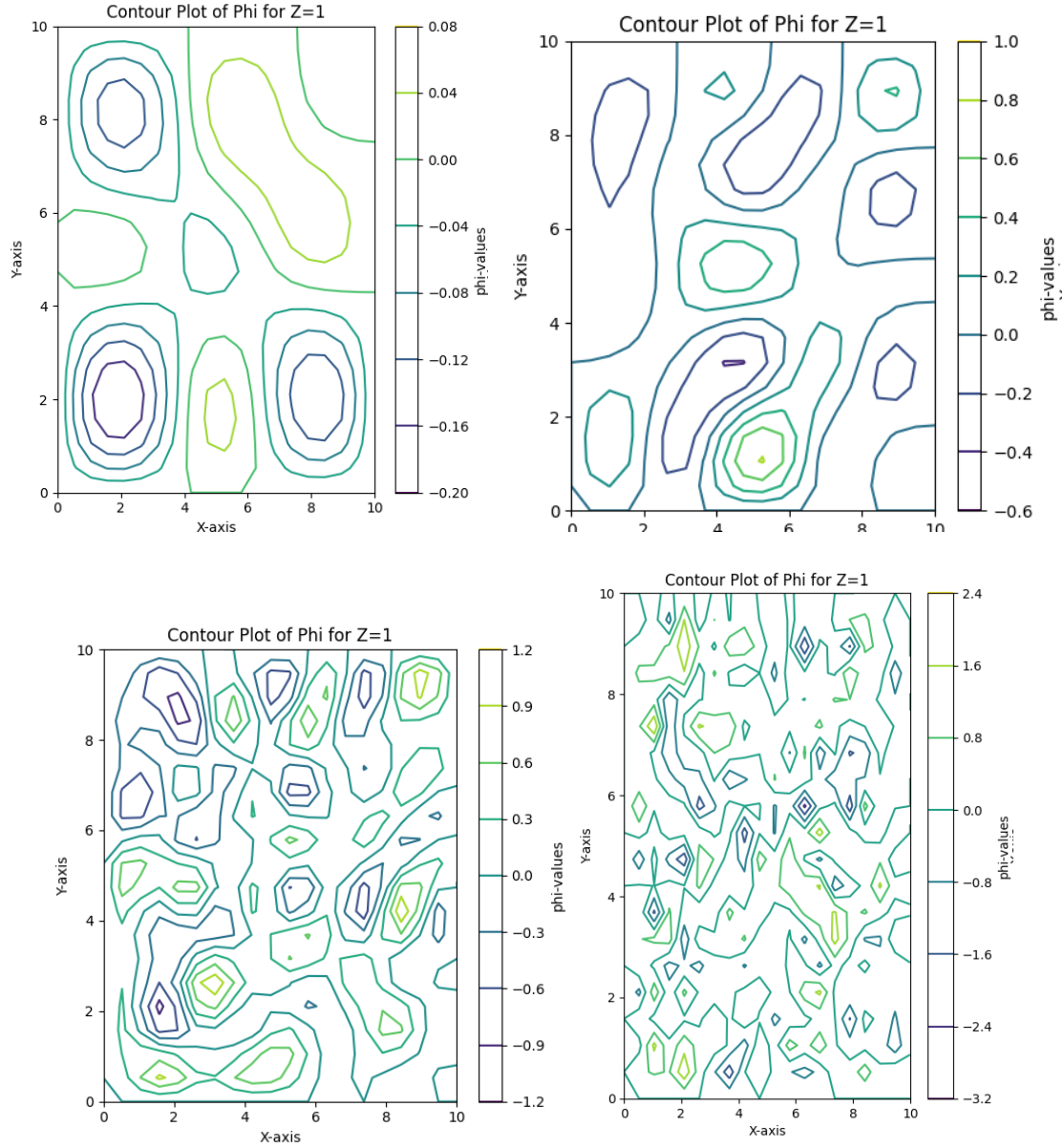


Figure 3: Plots of $\phi(x, y, 1)$ with $m=1$ and 27 unique modes (Left), 125 unique modes (Right), 1000 modes (Bottom Left) and 8000 modes (Bottom Right) for $\phi_{\vec{n}}$.

2 Dimensions Compared to 2-Dimensional Slicing

The 2-dimensional case for the fluctuation varies considerably and is not as simple as taking slices in one axis of $\phi(x, y, z)$. The main differences lie in the fact that we no longer have a 3-dimensional Fourier transformation, but a 2-dimensional one. This eliminates one of the sine functions and changes the exponent of the leading coefficient $2/L$ from $3/2$ to 1 . This missing sine function allows the amplitudes to be larger as the maximum of the absolute sine function is 1 . Furthermore, the frequency of each decoupled harmonic oscillator is in turn decreased, as ω_k depends on the length of the mode chosen. By removing one dimension, we've removed one of the terms squared to obtain the frequency of the harmonic oscillator, ultimately increasing the spread of the probability distribution function. As we can see from Figure 4, the fluctuations have an overall increase in amplitude for the 2-dimensional case compared to the slicing.

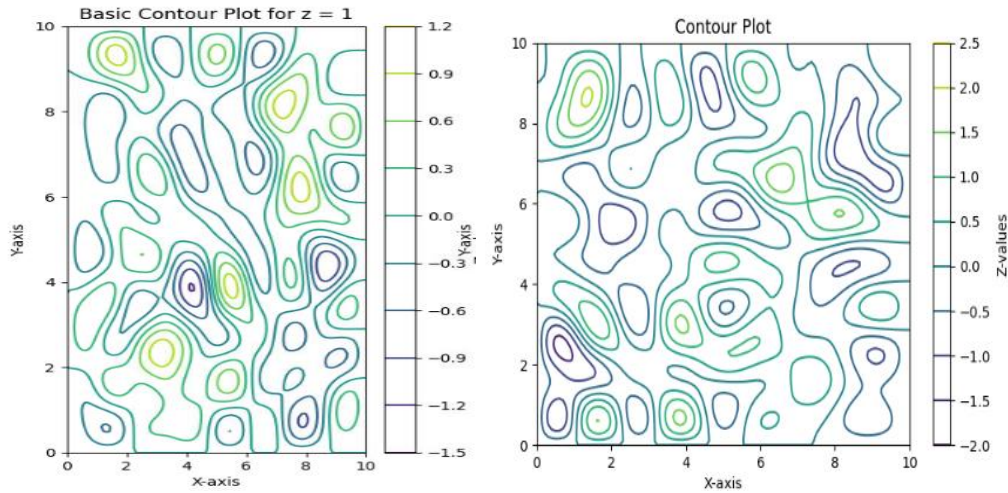


Figure 4: (Left) 2-dimensional slicing. (Right) 2-dimensional case. For each plot, the same parameters were used.

Wave Packets

Next, the wave packet case was considered, where a finite amount of decoupled harmonic oscillators are now excited and no longer in the ground state, specifically with one particle. As discussed in the Methods section, the probability distribution for the amplitudes $\phi_{\vec{n}}$ changes for the excited state, as larger amplitudes are more likely to occur than for the ground state. Therefore, we expect for the excited modes the amplitude of the fluctuations for $\phi(x, y, z)$ will increase, which

can be observed in Figure 5. Due to restrictions in the coding aspect, only excitations with 1 particle were considered which ultimately may have made it difficult to distinguish between the two plots below.

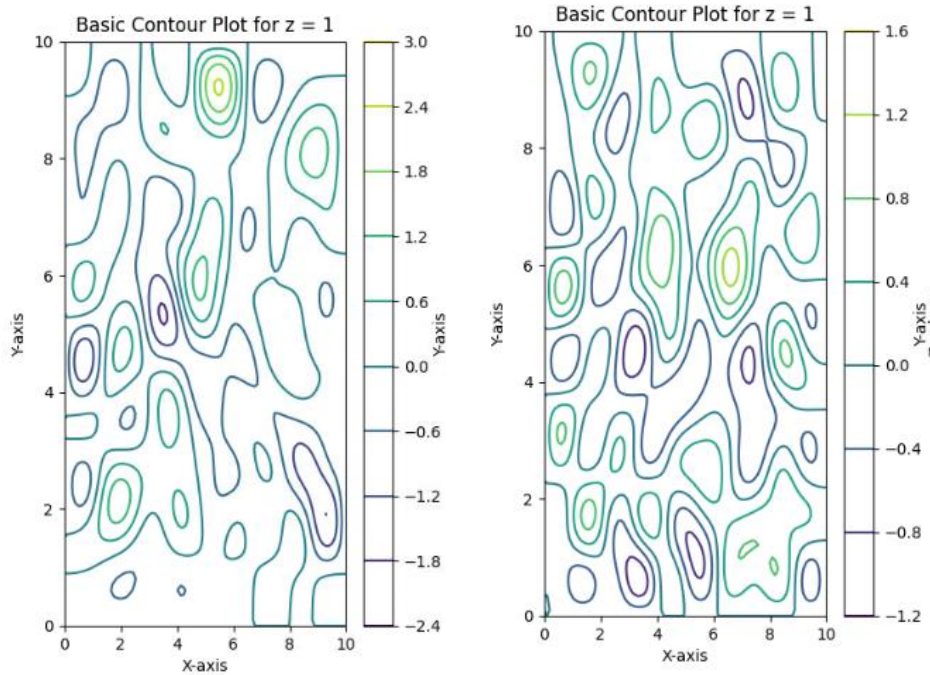


Figure 5: (Left) Wave packet state with finite excited modes ($m=1$) and ground state oscillators. (Right) Joint ground state. Identical parameters were used for both cases.

Suggestions for Further Research and Final Thoughts

Given the results of this project, further spacetimes should be considered outside of Minkowski space, such as FRW spacetimes where comoving coordinates are used instead.

Additionally, an instance that should be looked into is what happens when keeping the length of the mode functions the same as you increase the number of modes. To clarify, we would generate a finite amount of $\vec{n}$ where each vector has the same length. This would allow one to determine if increasing the amount of $\phi_{\vec{n}}$ coefficients used had more particular implications than what was described above. Lastly, the case for the wave packet state needs to be observed further, as there were few excitations presented in the calculations above, which can be seen through the program written as well. It would be interesting to witness what happens when $m=100$ for example for various harmonic oscillators.